RESEARCH PROPOSAL — EXTENSION

Graph-of-Loops Control for Continuous Autonomous Business Operation

An investigation into whether mutually-checking control loops, anchored in code-computed measurement, hard constraints and human judgement, can sustain a business operated by the department architecture we have already built and evaluated.

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This document describes the Orion architecture as specified. It extends an existing evaluated system; §3 states precisely what is already measured and what is not.

Abstract

We have built and evaluated a department-structured multi-agent architecture that generates a complete founding business plan from one sentence, using twenty-one agents across five departments over an enforced append-only state contract. Two measured findings from that work motivate this proposal. First, cross-agent constraints expressed as prompt instructions containing explicit numeric bounds were ignored repeatedly, and had to be re-implemented as post-conditions in code. Second, decomposing a quantitative estimate across dependent agents compounded variance: on financial accuracy the pipeline's standard deviation was 4.9× a single-LLM baseline's over repeated identical inputs, with a 7.6× best-to-worst spread against the baseline's 1.8×. Both findings indicate that an agent cannot be relied upon to assert a quantity by which it is judged. We propose to generalise the local remedy — derived figures override asserted ones — into a control architecture for *continuous* operation, in which every measure the business is judged by is computed by code from an append-only record and is unwritable by any agent, hard constraints are enforced in the execution path, and monitor loops are constrained by two topological invariants. We propose to evaluate this by extending our existing four-arm harness with four further arms, retaining all current metrics and adding five, and measuring proxy-anchor divergence over an operating horizon. We pre-commit to reporting null and refuting outcomes on the same terms as supporting ones.

1 Introduction and Motivation

Multi-agent systems built on large language models can decompose an objective, assign roles and produce coordinated output. Our own prior work establishes that such decomposition reliably improves coverage of a specification: a twenty-one-agent pipeline filled 100% of a twenty-six-field rubric where a single-LLM baseline reached roughly 81%.

That work also produced a result we did not expect and did not tune away. On the one metric that required a quantitative estimate to travel through a chain of dependent agents, the single-LLM baseline outperformed the pipeline — not by being more accurate at its best, but by being far more stable. The pipeline's best run was the more accurate of the two; its distribution was 4.9× wider. We characterise the architecture as unreliably better rather than reliably worse.

We suggest that this observation, together with the finding that prompt-level numeric constraints were ignored, points to something more general than a defect in our implementation. In both cases the system produced a number that served as its own evidence. When we removed that arrangement — computing the value from line items the agents had already produced rather than accepting the value they asserted — every one of five conflicts resolved.

The question this proposal takes up is whether that principle scales from a single contested field to an entire operating business. If a plan becomes a company's opening state rather than its final output, then the company acquires measures of its own performance, and the same problem recurs at a larger scale and over a longer horizon. We do not take the answer as established; it is the proposition motivating the work.

2 Problem Statement

> **CENTRAL QUESTION**
>
> Can a graph of mutually-checking control loops, anchored in code-computed measurement, hard constraints and human judgement, sustain the continuous autonomous operation of a business without proxy capture — under conditions where a single goal-measure-act loop per department does not?

Three sub-problems follow. A *measurement* problem: the quantities by which the system judges itself must be constructed so that it cannot influence them except by changing the world they describe. Our prior variance finding sharpens this — it is not sufficient that a measure be well-chosen if it is produced by the chain being

measured. A *topology* problem: adding oversight does not automatically add rigour, since components that check one another may converge on mutual agreement rather than accuracy, and the conditions distinguishing these cases require specification. And an *authority* problem: some judgements are not reducible to measurement, and the boundary between what is delegated and what is reserved must be explicit, enforced, and sustainable within a realistic human attention budget.

We note at the outset that we do not expect a positive result to show the elimination of proxy capture. The more plausible outcome, if the hypotheses hold at all, is that capture becomes detectable sooner and to a measurable degree.

3 What Is Already Established

This proposal extends an evaluated system rather than describing a new one. Stating the current evidentiary position precisely is necessary, because the proposed work reuses the same harness, arms and prompt set.

RQ	QUESTION	STATUS	POSITION
RQ1	Does a multi-agent pipeline beat a single-LLM baseline on accuracy, completeness and consistency?	Partly measured	Completeness favours decomposition decisively; consistency is effectively tied; financial accuracy favours the baseline
RQ2	Does dynamic routing improve efficiency without degrading quality?	**Supported**	17.3% token saving with a 0.0 pp completeness delta on prompts where a department was actually skipped
RQ3	Does grounding in retrieved data improve accuracy over parametric knowledge?	Not yet measurable	Requires hand-verified competitor ground truth, not yet collected; indirect signal only

Three qualifications apply to all of the above and are carried forward unchanged into this proposal. Present figures come from a five-prompt smoke run rather than the full thirty-five-prompt sweep. Financial accuracy divides by sector margin benchmarks that are not yet sourced, so its absolute values are provisional; between-arm comparisons are less affected because both arms divide by the same benchmark. And competitor coverage is unmeasured.

Two further findings are methodological rather than question-specific, and they are the load-bearing inputs to this proposal.

M1 — Prompt-level constraints were insufficient. A market-sizing prompt containing an explicit numeric sanity range, and a revenue prompt stating that year-one revenue must not exceed the obtainable market, were both ignored repeatedly. Five conflicts of this kind were identified, each corrupting the financial metric.

M2 — Variance compounds along a dependency chain. Over five runs of an identical prompt, the multi-agent arm produced MAPE of 290.9, 129.1, 239.2, 96.8 and 38.4 — standard deviation 103.8 against the baseline's 21.2 on the same prompt. The best pipeline run (38.4) beat the best baseline run (50.0). Five implementation defects were found and fixed before this conclusion was drawn, reducing MAPE from roughly 180–450 to roughly 70–200; we stopped there deliberately, on the grounds that further adjustment aimed at the metric would be tuning to the measurement.



Figure 1 — The measured basis for M2. All spread is run-to-run variance, since the input is identical across runs.

4 Related Work

Multi-agent frameworks and business tasks

MetaGPT [1] shows that role specialisation with structured intermediate artefacts outperforms free-form agent chat; our shared state object plays the role of its structured documents. ChatDev [2] reduces hallucination through inter-agent dialogue, where we reduce it by making certain quantities non-negotiable. AutoGen [3] is a capability layer rather than an architecture for a specific task. A recent survey [4] compares agentic frameworks on communication, memory and safety. BusiAgent [5] is the closest prior work in the business domain, targeting managerial decision support in established organisations rather than founding a new enterprise. None of these, in our reading, constrains what a unit may measure about itself, or treats the unit as a persistent locus of learning across sessions.

Durable execution and business process management

Workflow engines such as Temporal [6] provide restart-tolerant long-running execution with signals for external intervention; business process management systems such as Camunda [7] provide explicit approval gates, role assignment and audit trails. Both traditions address continuity and governance convincingly and we adopt their mechanisms. Neither, as far as we can determine, addresses whether a process's own success measures can be captured by that process — the metric is typically taken as given.

Reflection and organisational memory

Reflexion [8] establishes that an agent can improve by generating natural-language reflections on failure and retrieving them subsequently. Generative Agents [9] demonstrates memory streams with retrieval, higher-level reflection and planning over extended simulated periods. Both are important precedents for our memory and reflection design. In both, reflection is grounded in the agent's own record of events. Whether grounding reflection in externally computed measurement changes its behaviour is, in our reading, open, and it is among the questions we propose to examine.

Proxy capture

Goodhart's original observation [10], in Strathern's widely-cited formulation [11], holds that a measure ceases to be a good measure once it becomes a target. Manheim and Garrabrant [12] decompose the effect into distinct variants including regressional, extremal and causal forms. Amodei et al. [13] identify reward hacking among the concrete problems of AI safety, and Skalse et al. [14] give a formal definition. This literature characterises the failure thoroughly. Our reading is that it is less concerned with architectural remedies at the level of a multi-unit organisation, which is the gap this proposal addresses.

Convergent auditing and capability layers

A multi-lens convergent audit methodology [15] sweeps a product inventory through deliberately diverse lenses, verifying each candidate finding against the code before reporting, and terminating only after consecutive

passes yield nothing new. Its claim-versus-code lens is the conceptual origin of the claim-versus-record check proposed in §8. That project is explicit that its convergence is a heuristic bound rather than a proof and that its runs are not deterministic; we adopt both the technique and that candour. A separate capability layer [16] provides external source access through ordered preferred-and-fallback backends per channel with health diagnostics, and informs our tool registry. We restrict use to its zero-configuration channels and exclude session-credential social scraping, for the reasons in §15.

5 Problem Taxonomy

We propose to study four failure modes of continuous operation. For each: a definition, the mechanism we believe produces it, an instantiation in our domain, and — most importantly for the design of the study — how it might be measured.

MODE	MECHANISM	INSTANTIATION IN OUR DOMAIN	PROPOSED MEASUREMENT
F1 Proxy capture	The evaluated quantity correlates with the objective under ordinary conditions; optimisation selects for actions that decouple them	Marketing evaluated on footfall runs discounts deep enough that each additional customer reduces net margin	Rank correlation between a proxy series and its corresponding anchor series over a sliding window
F2 Objective non-interrogation	The objective is exogenous to the loop and is never a candidate for revision	"Increase footfall" persists after marginal footfall becomes loss-making	Elapsed operating time during which an objective holds while its intended anchor moves adversely; revision lag
F3 Inter-department conflict	Objectives are per department; nothing is accountable for their interaction	Finance trims the supplier budget that would have addressed the churn Marketing's discounting produced	Occurrences where two departments' own measures improve while a shared anchor moves adversely; detection latency
F4 Measurement decay	A stale value returns a number rather than an error, so staleness produces no signal	A margin benchmark set at founding still classifies performance as acceptable after sector economics shift	Metric age at time of use; retrieval-weighted memory age; interval from an injected perturbation to threshold revision

F1 and F3 are the modes our existing consistency metric gestures at without capturing. That metric checks agreement between departments within a single generated plan. It has no temporal dimension, and it compares departments against each other rather than against an external quantity — which is precisely the arrangement §2 identifies as insufficient.

6 Research Questions

RQ1–RQ3 are retained unchanged, with the status recorded in §3. RQ4–RQ7 are proposed. The numbering is continuous so that the two bodies of work read as one programme.

RQ4 Does variance compounding persist when the estimated quantity is replaced by a quantity computed from a record of events?

> This connects the proposed architecture directly to M2. It is the question we consider most likely to yield a clean answer, because the comparison is against an already-measured baseline.

RQ5 Does proxy-anchor divergence arise systematically under single-loop control in a continuously operated business, and if so, how quickly?

> Establishes whether the phenomenon is present and measurable in this setting at all. If it does not arise, RQ6 and RQ7 are moot and we would report that.

RQ6 Do independent monitor loops detect divergence earlier than a composite business-health indicator?

> Compares detection latency between monitors reading anchors and a conventional aggregate health view.

RQ7 **Are the proposed graph invariants — cycle termination at an anchor or human, and monitor independence — necessary for detection, or merely sufficient?**

> Examined by relaxing each invariant individually. We regard this as the most theoretically interesting question and the least likely to yield a clean answer, since relaxing a structural invariant may change several things at once.

A fifth question concerns feasibility rather than mechanism, and we treat it as a first-class object of study rather than an assumption: **what human decision load does the governance arrangement impose, and does it remain within a plausible attention budget?** A governance design that produces more decisions than a person would genuinely review is not viable however sound its logic.

7 Hypotheses and Falsification Conditions

Each hypothesis is stated so that a specific observation would refute it. We commit to these conditions before running the arms, and to reporting the outcome under them whatever it is.

H1 We hypothesise that **a margin computed from a record of transactions will exhibit materially lower run-to-run variance than the same margin estimated through the agent dependency chain, on identical inputs.**

> **Would be refuted if:** the computed quantity's standard deviation over repeated identical runs is not materially below the 103.8 already measured for the estimated quantity.

H2 We hypothesise that **under single-loop control with proxy metrics, proxy and anchor series will diverge systematically over an operating horizon, with proxies improving while anchors deteriorate.**

> **Would be refuted if:** the two series remain positively rank-correlated across the majority of seeded runs, or divergence appears without consistent direction — which would indicate noise rather than capture.

H3 We hypothesise that **under the full loop-graph topology, terminal proxy-anchor divergence will be materially smaller than under single-loop control, given identical seeds, models and starting state.**

> **Would be refuted if:** terminal divergence under the full topology is statistically indistinguishable from the single-loop arm, or larger.

H4 We hypothesise that **monitor loops reading anchors will detect a synthetically induced capture scenario earlier than a composite business-health indicator.**

> **Would be refuted if:** monitor detection latency is equal to or greater than that of the composite indicator across induced scenarios.

H5 We hypothesise that **relaxing cycle-termination — permitting cycles composed only of agents — will degrade detection of induced capture relative to the full topology.**

> **Would be refuted if:** detection performance is unchanged when the invariant is relaxed, which would indicate the invariant is not doing the work we attribute to it.

H6 We hypothesise that **department-scoped memory will retain retrieval precision as the corpus grows across operating cycles, whereas a shared context will degrade.**

> **Would be refuted if:** precision degrades at a comparable rate under both arrangements, or shared context matches scoped memory at the largest corpus sizes tested.

> **EXPECTED STRENGTH, STATED IN ADVANCE**
>
> We regard H1 and H6 as the most likely to be supported, H2 as likely but possibly slow to appear within a bounded horizon, H3 and H4 as genuinely uncertain, and H5 as the most likely to yield an ambiguous outcome. We record these expectations now so that they can be compared against what we find.

8 Proposed Architecture as Intervention

The architecture below is the independent variable. Its value to this work is that it can be ablated, and that three of its four components already exist in a measured form.

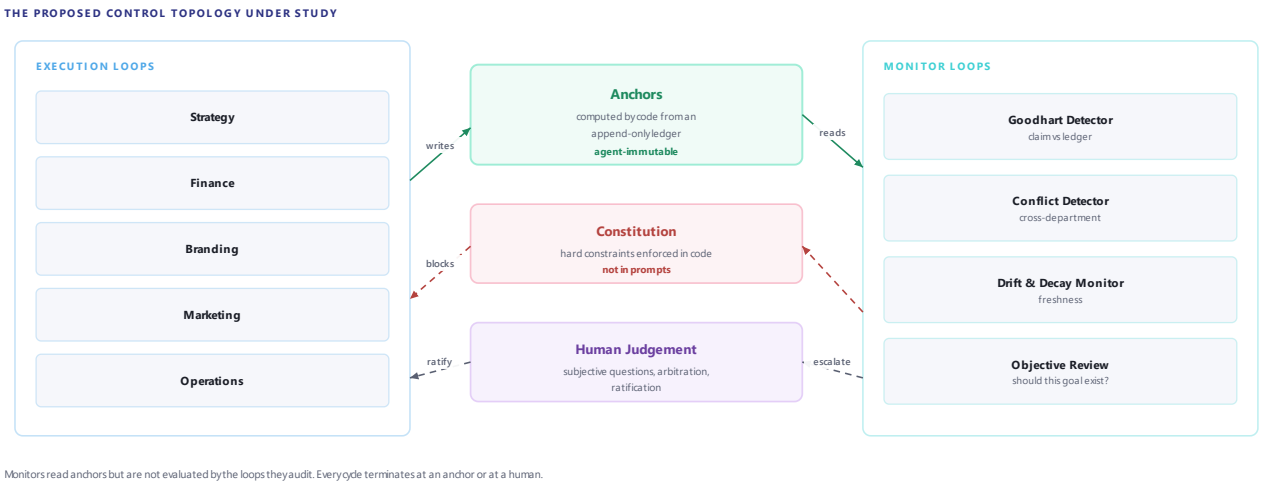


Figure 2 — The proposed control topology. The five execution loops are the departments already implemented; the four monitor loops and the anchor layer are the proposed addition.

Anchors — a generalisation of an existing mechanism

Our current system already computes rather than accepts certain quantities: cost totals are summed from line items, operating expense is reconciled against that sum, margins and break-even are computed in Python from six requested drivers. The proposal generalises this to every quantity the business is judged by, derived from an append-only record of events, with the invariant that no component of the agent runtime holds a write path to an anchor and no learning process optimises directly against one. The intended analogy is the held-out evaluation set, whose value as a measure is conventionally attributed to its inaccessibility to the training process.

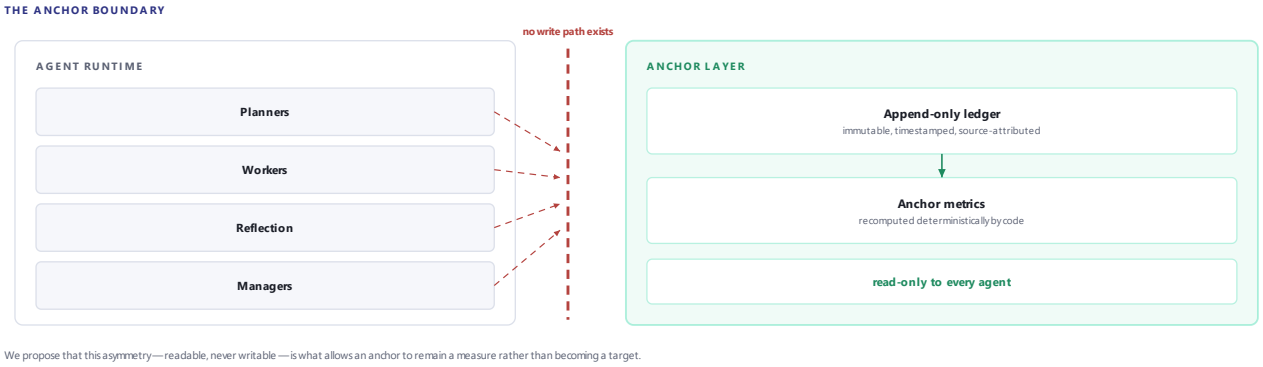


Figure 3 — The proposed asymmetry: readable by every component, writable by none. We propose that this asymmetry, rather than any property of the metrics themselves, is what allows them to remain measures.

Constitution — a generalisation of M1

M1 established that numeric bounds stated in prompts were ignored, and that the same bounds enforced as post-conditions held. The Constitution applies that result to operating constraints — spend caps, irreversibility, data deletion — enforced at the execution path rather than described in prompts. Whether this produces a measurable difference in constraint violation rates under continuous operation is a subsidiary question we intend to record.

Graph invariants

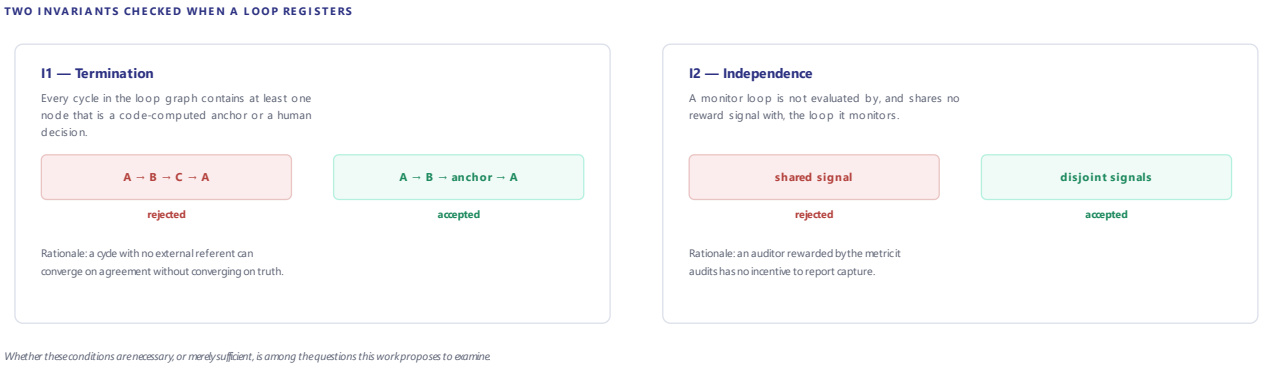


Figure 4 — The two proposed invariants and the rationale for each. Whether they are necessary, or only sufficient, is the subject of RQ7.

The motivation for the first invariant is that a set of components which evaluate one another may reach agreement without reference to any external quantity. We describe such a configuration as a larger single loop rather than a graph of loops. The invariant attempts to state formally what distinguishes the two, and we acknowledge that our formulation may prove too strong, too weak, or the wrong frame entirely.

9 Methodology

We propose to extend the existing harness rather than replace it. The four current arms are retained unmodified, so results already obtained remain valid and comparable; four arms are added for the operating layer.

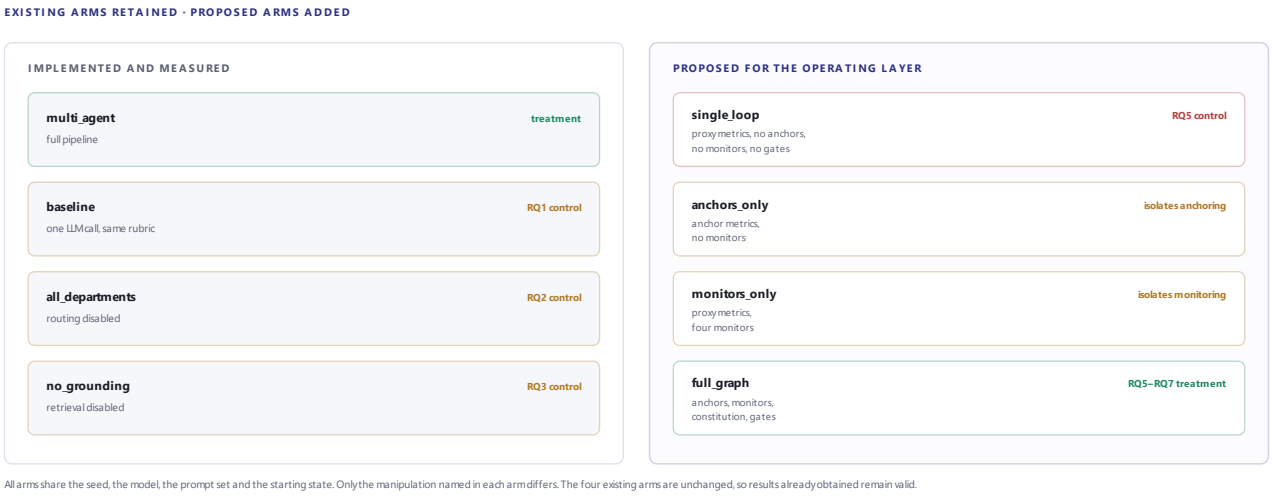


Figure 5 — Existing arms retained; proposed arms added. All arms share the seed, the model, the prompt set and the starting state.

Baseline fairness

The credibility of the comparison rests on the single-loop arm being a fair representative of ordinary practice rather than a weakened straw comparison. Our existing baseline discipline is instructive here and is carried forward: the current single-LLM control receives the stronger model, an explicit enumeration of every rubric field, and the same deterministic intent classifier, and is pinned to one named provider after an earlier implementation silently served a "Gemini baseline" from a local three-billion-parameter model.

Applying the same discipline: the single-loop arm will receive the same memory, planning and execution machinery as the full-graph arm, differing only in control topology; its proxy metrics will be drawn from measures actually used in comparable business functions rather than chosen for exploitability; and its specification will be fixed and documented before any arm is run. We regard this as the principal

methodological risk of the study, and we would rather report a null result against a strong baseline than a positive one against a weak baseline.

Environment

Operation requires an environment that responds to the company's actions. We propose a stochastic model in which customers arrive, convert, retain or churn according to parameters known to the harness and not to the agents. This is what makes anchors computable and the study possible, and it imposes the external-validity limit discussed in §12. The prompt set, gazetteer and sector structure are reused unchanged.

Induced scenarios

Two families of perturbation at predetermined points. *Capture scenarios* make a proxy improvable by an action that damages the corresponding anchor, allowing detection to be timed against a known onset. *Decay scenarios* shift environment parameters so that calibrated thresholds silently lose validity. Onset times are known to the harness, which converts detection into a measurable latency rather than a judgement.

10 Metrics and Instrumentation

The five existing metrics are retained with their definitions unchanged. Five are proposed.

ID	METRIC	DEFINITION	STATUS
F54	Financial accuracy	MAPE of projected steady-state net margin against sector benchmark	Existing — provisional benchmarks
F55	Competitor coverage	Verified competitors found over hand-collected ground truth within 2 km	Existing — not yet measurable
F56	Plan completeness	Expected fields present and substantive over a 26-field rubric	**Existing — measured**
F57	Consistency	Proportion of applicable cross-department checks passing	**Existing — measured**
F58	Routing efficiency	Token and time saving against the forced-all control, with completeness delta	**Existing — measured**
F59	**Proxy-anchor divergence**	Rank correlation between a proxy series and its corresponding anchor series over a sliding window	Proposed
F60	**Detection latency**	Operating time from the known onset of an induced scenario to the first finding citing the affected anchor	Proposed
F61	**Conflict incident rate**	Occurrences where two departments' measures improve while a shared anchor moves adversely	Proposed
F62	**Objective staleness**	Time an objective holds while its intended anchor moves adversely; revision lag	Proposed
F63	**Human decision load**	Decisions escalated per operating day and their severity distribution	Proposed

Every proposed metric is computed by the harness from the append-only record. None is derived from any agent's account of its own performance. We consider this a methodological requirement rather than an implementation preference: if the instrument for detecting proxy capture could itself be influenced by the departments under study, the study would exhibit the failure it proposes to measure. This is the point on which we would most welcome scrutiny.

The existing refusal-to-impute policy is retained. A metric that cannot be computed records `applicable=false` with a reason rather than a zero, and the generated report lists all unscored measurements grouped by reason.

We note that this policy is what currently prevents F55 from being reported as a misleading figure, and we regard it as more valuable than any single result it withholds.

11 Experimental Protocol

Episode structure. An episode is a fixed number of operating cycles over one seeded company, initialised from a generated plan. Each arm runs on an identical seed set, so comparisons are paired rather than between independent samples.

Repetition. Because model behaviour is non-deterministic — the point M2 makes quantitatively — each seed is run repeatedly and per-seed distributions rather than point values are reported. The repetition count will be fixed after a pilot establishes run-to-run variance, following the same practice that produced M2.

Pre-registration. Hypotheses, falsification conditions, metric definitions, the single-loop arm specification and the analysis plan will be recorded and timestamped before the arms are run. Any deviation will be reported as a deviation rather than presented as original intent.

Artefact. The existing artefact runs entirely on free-tier infrastructure with a test suite that provably consumes no API quota, enforced by a socket guard. That property is retained. Harness, seeds, record exports and analysis scripts are intended for release so that reported figures can be recomputed from raw records by a third party.

Reporting. Aggregate results will be accompanied by per-seed traces, including runs that behave contrary to the aggregate. Given non-determinism the distribution is more informative than the mean, and selective presentation of favourable runs would be the most likely way for this study to mislead.

Outstanding prerequisites. Three items from the existing work must be closed before the proposed arms can be interpreted: the full thirty-five-prompt sweep, sourced sector benchmarks, and collected competitor ground truth. Until they are, F54 remains provisional and F55 remains unmeasurable, and any operating-layer result resting on them inherits that status.

12 Threats to Validity

Construct validity

The central construct is that proxy-anchor divergence captures what Goodhart's Law describes. This may be too coarse. Divergence can arise from ordinary lag between an action and its effect, from mis-specification of which anchor a proxy serves, or from noise — and M2 establishes that noise in this system is substantial. Manheim and Garrabrant's decomposition [12] suggests several mechanistically distinct effects would be conflated by a single divergence measure. We propose to require directionality and persistence rather than any single-window divergence, and to report variants separately where instrumentation permits. We do not expect to resolve this fully, and we regard it as the most serious threat.

Internal validity

Non-determinism is a confound: an observed difference between arms may reflect sampling rather than topology. Paired seeds, repetition and distributional reporting are the mitigations, and M2 gives us a calibrated sense of the variance to expect. A second concern is that arms differ in more than the nominal variable — the full topology performs additional inference, so any advantage might be attributable to additional computation rather than structure. We propose a compute-matched control in which the single-loop arm receives equivalent inference budget spent within its own loop, and we note that designing this fairly is not straightforward.

External validity

The operating environment is simulated. Anchors would be computed honestly from a record of what the system did, but a simulated customer's decision to churn is a modelling assumption rather than an observation. Findings would therefore speak to the behaviour of control topologies under a specified environment model. A further limit is inherited: the existing prompt set is Indian, mostly Mumbai, in English, and the gazetteer, funding schemes, salary bands and benchmarks are all India-specific.

Reflexivity

The most uncomfortable threat is that the same author designs the architecture, selects the anchors, defines divergence, specifies the baseline and interprets the results. No amount of procedural care fully removes this. Pre-registration, artefact release and a documented baseline specification are partial mitigations; independent replication would be stronger, and we regard the artefact release as primarily serving that end rather than serving the presentation of these results.

Weak ground truth

Inherited and unresolved. Work on code-generating agents can execute unit tests; business quality has no comparable oracle. Our metrics are proxies — which is an uncomfortable observation for a study about proxy capture, and we state it rather than leave it implicit.

13 Anticipated Contributions

Stated conditionally, since they depend on outcomes not yet obtained. *If* the hypotheses are supported, the contributions would be:

1. **A loop-graph formalism** for multi-unit autonomous operation, in which loops are first-class objects with declared inputs, outputs, cadence and evaluation signal, and topology is validated rather than assumed.
2. **The anchor-immutability invariant** — that an organisation's measure of itself should be structurally unwritable by that organisation — generalising a mechanism we have already found necessary at the scale of a single contested field.
3. **Two topology constraints**, cycle termination and monitor independence, offered as a formal statement of what distinguishes a graph of mutually-checking loops from one larger loop.
4. **An extension of variance compounding to the operating regime**, and evidence on whether computing a quantity from a record rather than estimating it through a chain removes the effect.
5. **A reproducible testbed** for studying proxy capture in autonomous multi-unit systems, released with seeds and records.

Contribution 4 does not depend on the loop graph working. If H1 is supported and H3 is not, the study still establishes something useful about where decomposition variance can and cannot be removed.

14 Expected Outcomes, Including Negative Results

Three outcomes are possible and we commit in advance to reporting whichever obtains, on the same terms. The existing work provides a precedent: its central finding is a negative one, reported rather than tuned away.

OUTCOME	WHAT WE WOULD OBSERVE	HOW WE WOULD REPORT IT
Supporting	H2 supported; terminal divergence materially lower in the full-graph arm on paired seeds; monitors detect induced capture earlier than the composite indicator	As evidence that control topology affects proxy capture in this setting, bounded explicitly by the simulated environment and the scale studied
Null	Divergence arises but arms are indistinguishable; or divergence does not arise systematically within the horizon	As a null result, with analysis of whether the horizon was too short, the perturbations too weak, or the construct wrong — without recasting the study as something it was not
Refuting	The full-graph arm shows equal or greater divergence, or its monitors detect systematically later	As a refutation of H3 and H4, with an account of the likely mechanism. Oversight loops adding cost without adding detection would be a substantive finding in itself

A COMMITMENT

The architecture is the object of study, not a position to be defended. A refuting result would be reported with the same prominence as a supporting one, and the pre-registered falsification conditions in §7 exist so that this commitment can be checked rather than merely asserted.

15 Ethics, Safety and Data Handling

No real-world financial action. The system operates a simulated business. It holds no payment credentials and executes no real transaction.

External data and prompt injection. Retrieved content is untrusted input and may contain text addressed to the model. It is tagged with provenance, delimited, denied tool authority, and structurally prevented from moving an anchor or satisfying a hard constraint. The anchor invariant limits the consequence of a successful injection rather than guaranteeing against it. We propose to include adversarial injection cases as explicit test scenarios and to report failures.

Sources excluded on principle. External access is restricted to zero-configuration public sources. We exclude session-credential scraping of social platforms, which would require storing personal authentication state and would raise terms-of-service and personal-data questions disproportionate to the value added. This is a deliberate limit on the evidence base, taken knowingly. We note a related finding already recorded: OpenStreetMap proved a usable substitute for existence checking and not for price positioning, publishing no ratings and omitting sector tags that are common in the Indian market.

Human oversight. Questions classified as subjective — ethical acceptability, strategic direction, brand and tone — are routed to a human by rule rather than at an agent's discretion, and the sustainability of that load is treated as an object of study.

Personal data. Simulated customers are synthetic. No personal data is collected, and released artefacts contain none.

Third-party code. Open-source components are used under their stated licences, pinned to specific commits, and cited for architecture and licensing verified by direct inspection rather than for popularity metrics.

16 Timeline, Feasibility and Prerequisites



Figure 6 — Milestones with two decision gates. The gates protect the experiment: an ablation run on an unstable substrate would produce uninterpretable results.

The two gates are consequential. If continuous operation is not stable by the first, the proposed arms cannot be run meaningfully and the work would be re-scoped to a narrower claim — RQ4, which concerns variance and does not require sustained operation, is the natural fallback. If the harness is not producing reproducible paired runs by the second, reporting would reduce to a characterisation study addressing RQ5 alone.

We state these fallbacks explicitly because the most common failure of a proposal on this timeline is not that the system does not work, but that a weak experiment is run on schedule and reported as though it were a strong one.

Feasibility under local inference

The work uses hosted free tiers with a local terminal fallback, which constrains throughput and therefore the number of episodes. We argue this is adequate for the question: the hypotheses concern control structure, and all arms use the same provider chain, so differences between arms are attributable to topology rather than capability. A weaker model may in fact be a more sensitive instrument for F1, being less likely to notice unaided that a proxy has decoupled from its objective.

Two limitations follow and we would state them as such. Absolute business performance across all arms will be below what a frontier model would achieve, so terminal figures are not comparable to any commercial benchmark. And whether a substantially more capable model exhibits F1 to the same degree is not something this study can determine — a question noted in §17. We also carry forward an existing reproducibility hazard: a model named in our original design was retired during development, so we report the models actually used rather than those intended.

17 Future Work

Capability scaling. Whether proxy capture attenuates or intensifies with model capability, examined by holding topology fixed and varying the model. We consider this the most consequential question this work raises without answering.

Variance reduction by ensembling. M2 suggests ensembling the estimation chain rather than individual agents. This is orthogonal to the loop graph and would compose with it.

Real-market operation. Replacing simulated anchors with observed ones, under narrow scope and full human authority over irreversible action, would address the external-validity limit the simulation imposes.

Learned topology. The loop graph is hand-specified. Because it is a first-class validated object, a system that proposes its own monitors while remaining subject to the same invariants is a natural extension — and raises the question of whether a system can be trusted to design its own oversight.

Additional departments and geographies. Sales, HR and Legal are absent from the current five and would extend the conflict surface that F61 measures. Extension beyond a single geography is a prerequisite for any general claim.

Formal verification of the invariants. Cycle termination and monitor independence are graph properties, checked per instance at registration. Proving them for classes of topology appears tractable.

A qualitative track. An LLM-as-judge or human-rated assessment reported separately from the quantitative core, so that the reproducible part stays reproducible while persuasiveness and practical usefulness become visible.

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Citation status. References 8–14 were verified against their published records during preparation of this document. References 1–5 are carried forward from the existing work, where they were verified to exist with the titles and identifiers given; author lists and venues should be confirmed from the arXiv listing pages before submission. References 6–7 are project documentation, cited for capability descriptions rather than empirical claims. References 15–16 are open-source repositories, cited for architecture and licence terms verified by direct inspection. Citations for LangGraph, the model families actually used, and the sector benchmark sources remain outstanding.